Supplementary Material for Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures

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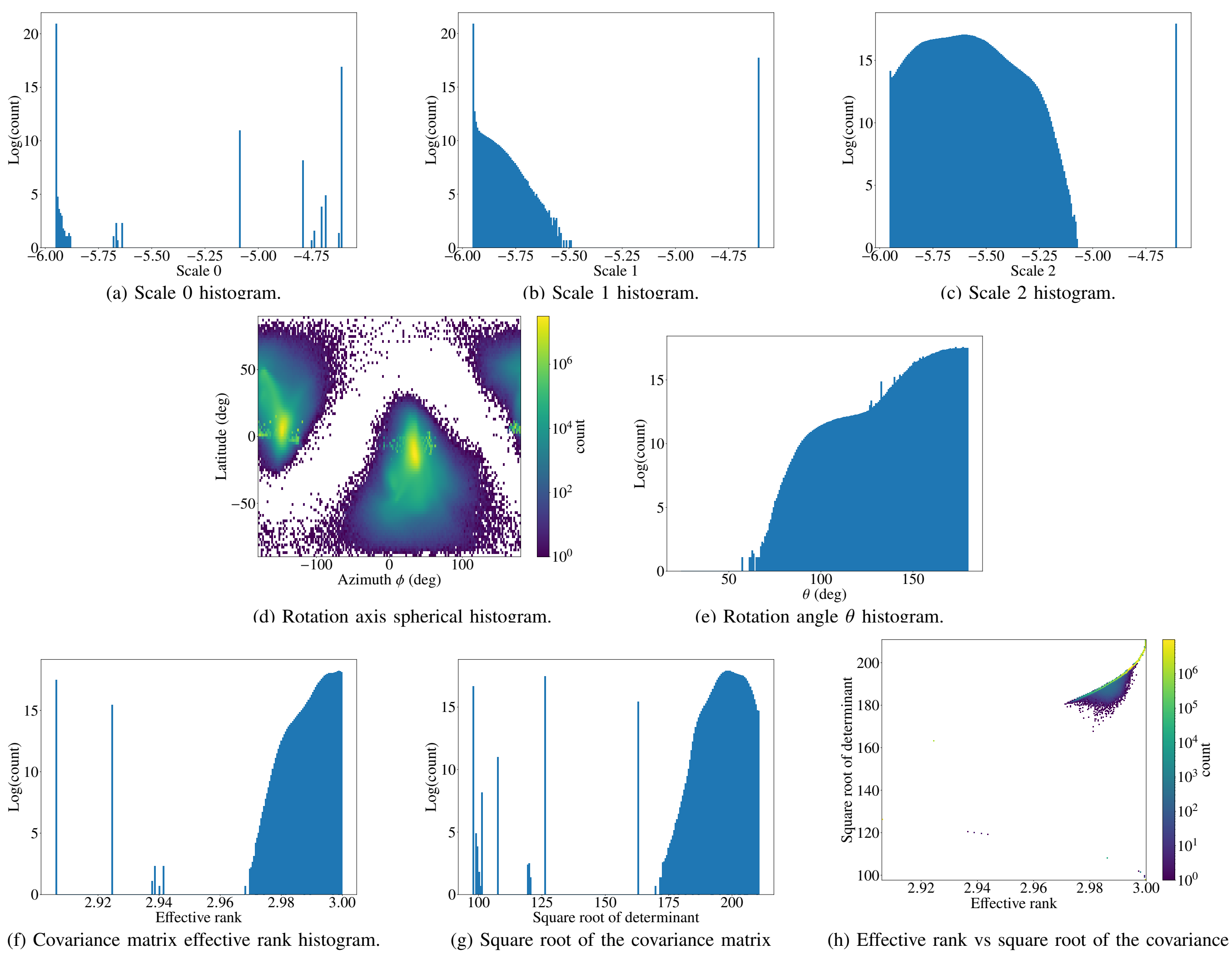
This document presents the supplemental material of the paper *Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures* (S. Croci *et al.*).

First, there is the analysis of the dynamic Gaussian splat (GS) reconstructions from the dataset mentioned in the paper. Second, there is the analysis of static reconstructions generated with a scene GS optimization method called Postshot from another paper *Gaussian Splatting vs. Classical Photogrammetry: A Comparison for Virtual Backdrops* (P. Haslbauer *et al.*).

I. Gaussian Splat Analysis of Dynamic Reconstructions

A. All Dynamic Reconstructions



(a) Scale 0 histogram. (b) Scale 1 histogram. (c) Scale 2 histogram.

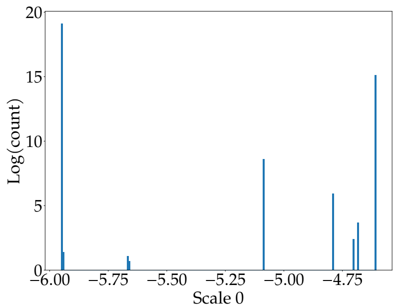
(d) Rotation axis spherical histogram. (e) Rotation angle θ histogram.

(f) Covariance matrix effective rank histogram. (g) Square root of the covariance matrix determinant. (h) Effective rank vs square root of the covariance matrix determinant.

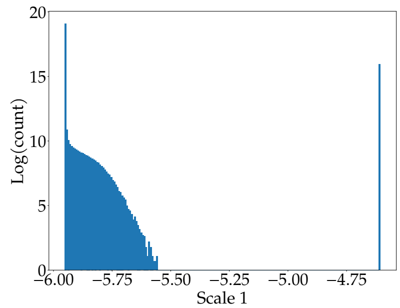
Fig. 1: Analysis of the Gaussian splats from all dynamic reconstructions.

This work was supported by the European Commission Horizon programme (GA 101135637, HEAT https://heat-xr.eu/).

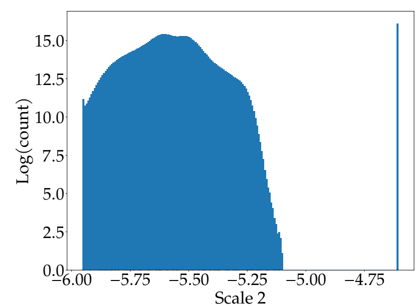
B. Den's Dynamic Reconstruction



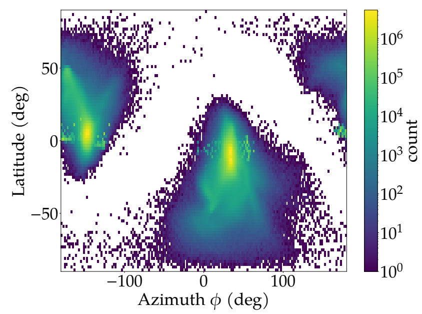
(a) Scale 0 histogram.



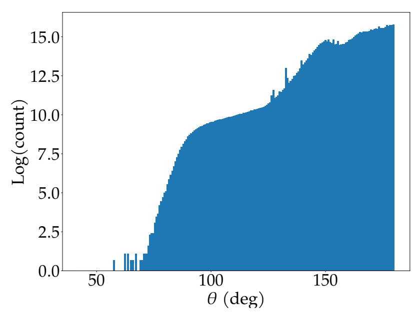
(b) Scale 1 histogram.



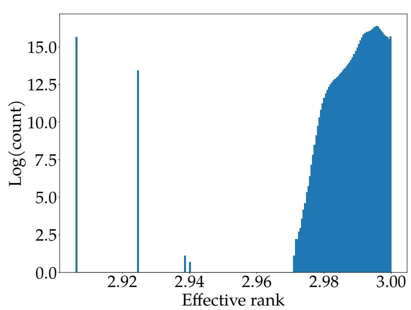
(c) Scale 2 histogram.



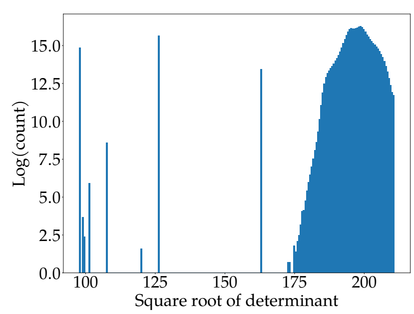
(d) Rotation axis spherical histogram.



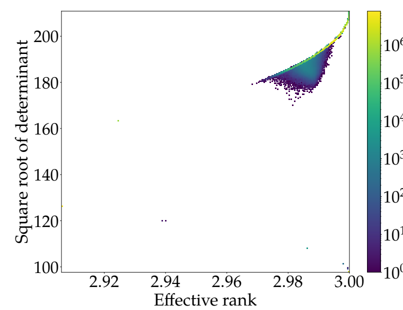
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



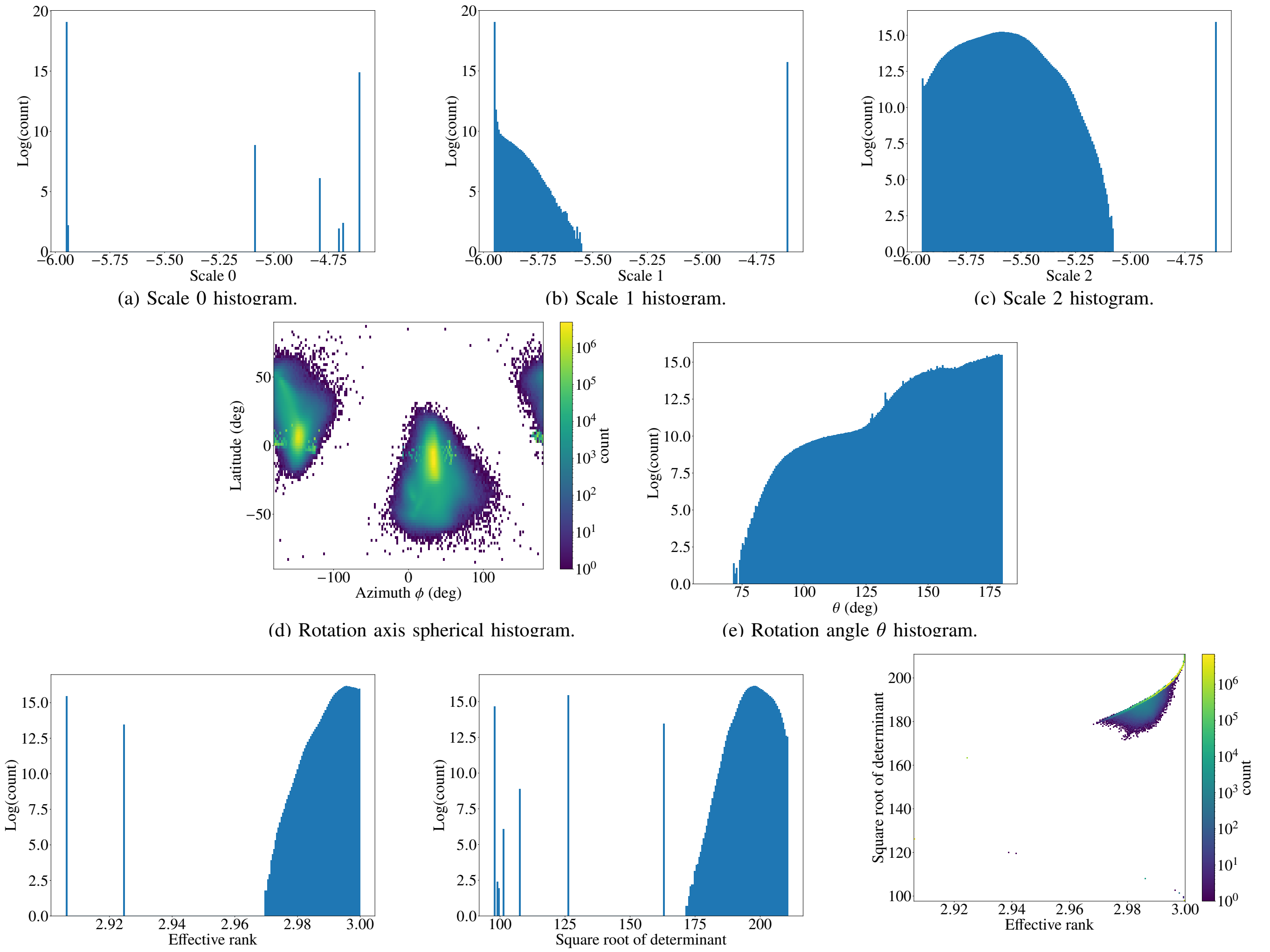
(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 2: Analysis of the Gaussian splats from Den's dynamic reconstruction.

C. Nathalie's Dynamic Reconstruction



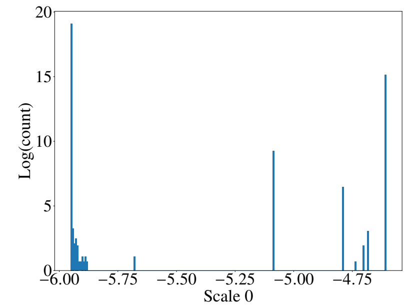
(a) Scale 0 histogram. (b) Scale 1 histogram. (c) Scale 2 histogram.

(d) Rotation axis spherical histogram. (e) Rotation angle θ histogram.

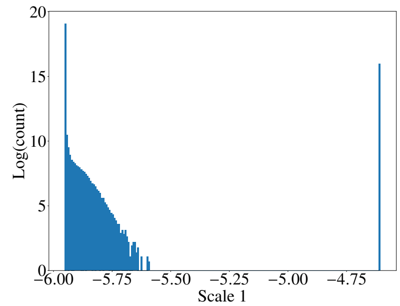
(f) Covariance matrix effective rank histogram. (g) Square root of the covariance matrix determinant. (h) Effective rank vs square root of the covariance matrix determinant.

Fig. 3: Analysis of the Gaussian splats from Nathalie's dynamic reconstruction.

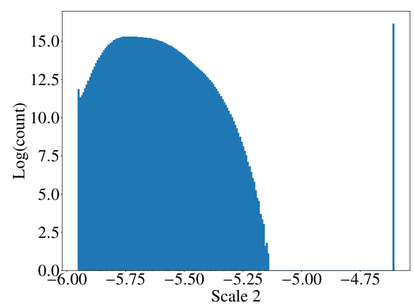
D. *Philipp's First Dynamic Reconstruction*



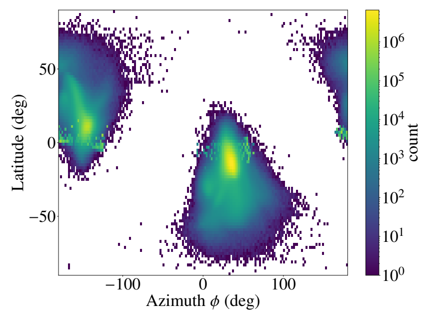
(a) Scale 0 histogram.



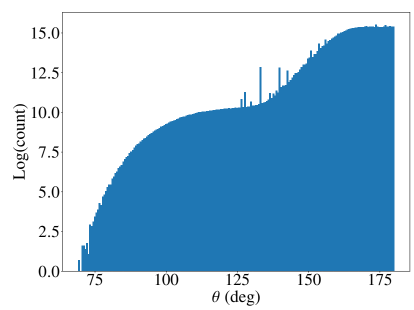
(b) Scale 1 histogram.



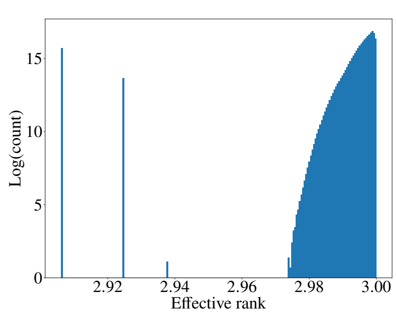
(c) Scale 2 histogram.



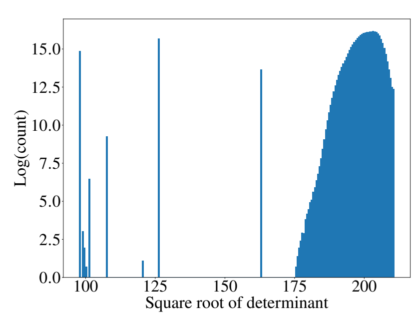
(d) Rotation axis spherical histogram.



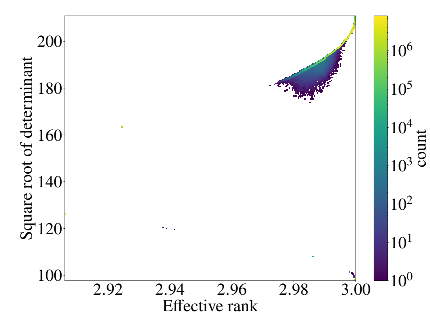
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



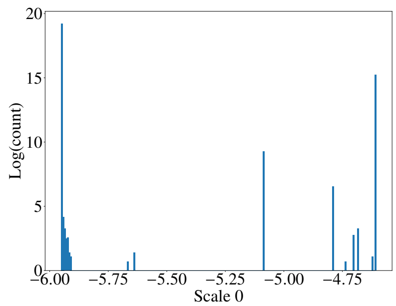
(g) Square root of the covariance matrix determinant.



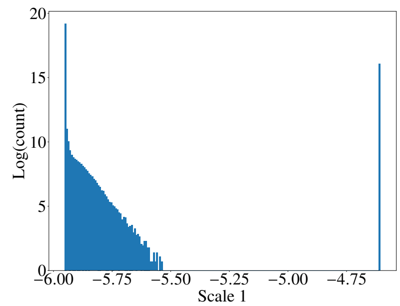
(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 4: Analysis of the Gaussian splats from Philipp's first dynamic reconstruction.

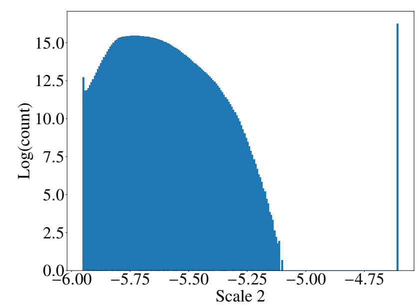
E. *Philipp's Second Dynamic Reconstruction*



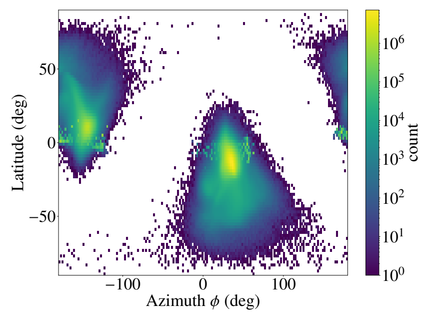
(a) Scale 0 histogram.



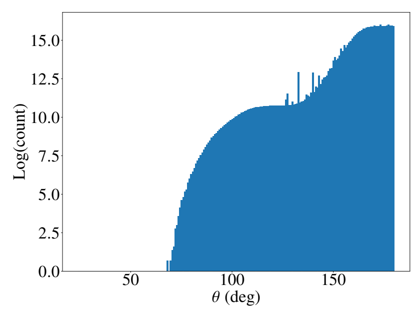
(b) Scale 1 histogram.



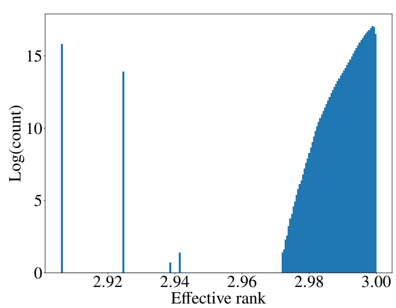
(c) Scale 2 histogram.



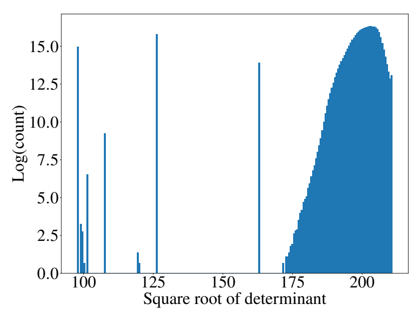
(d) Rotation axis spherical histogram.



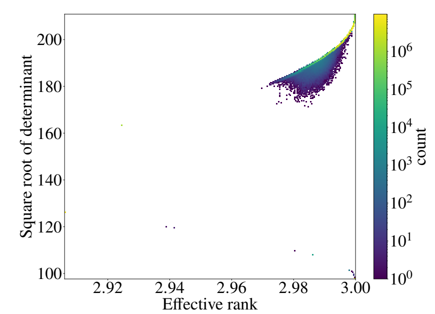
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



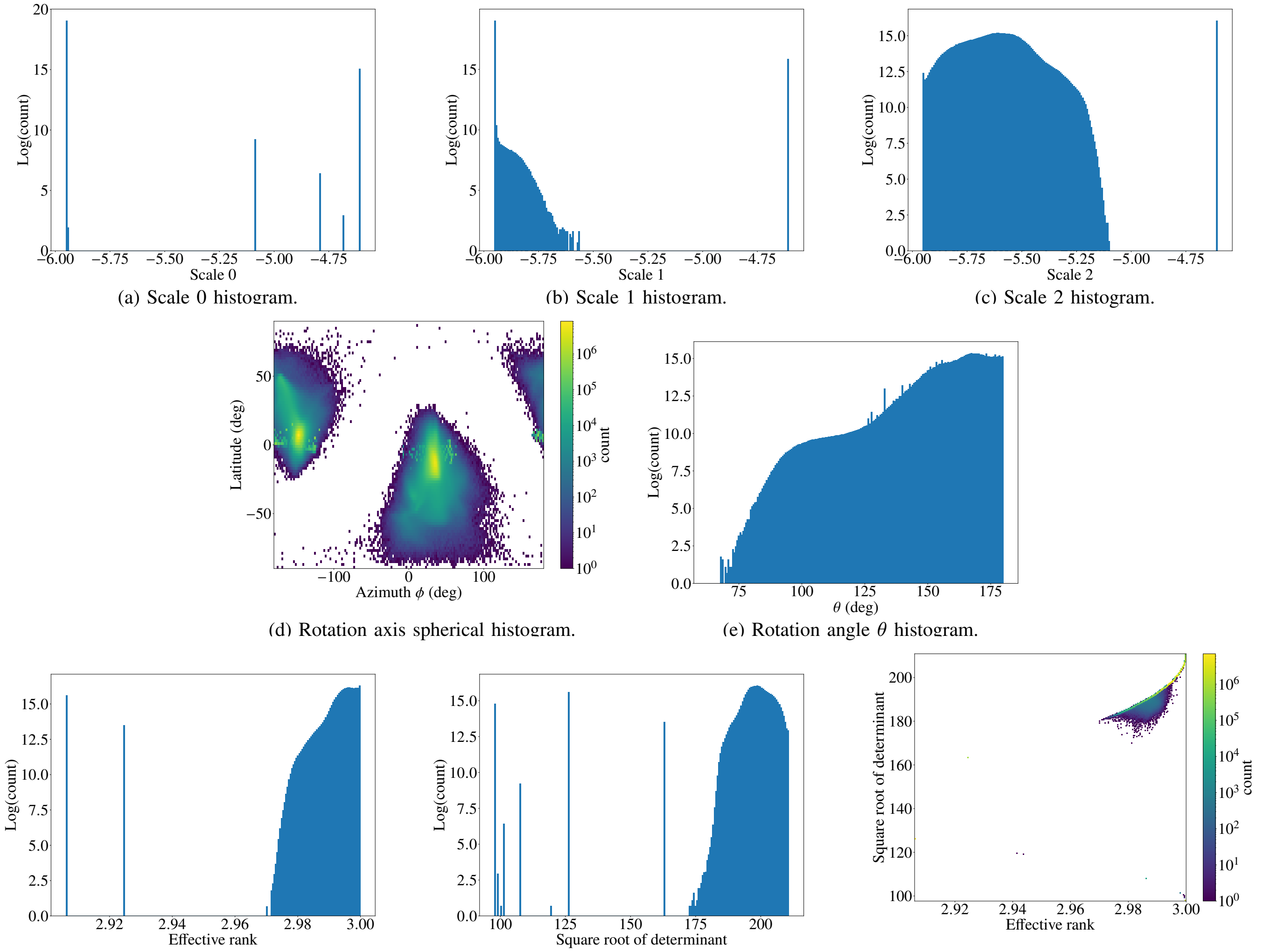
(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 5: Analysis of the Gaussian splats from Philipp's second dynamic reconstruction.

F. Simone's Dynamic Reconstruction



(a) Scale 0 histogram.

(b) Scale 1 histogram.

(c) Scale 2 histogram.

(d) Rotation axis spherical histogram.

(e) Rotation angle θ histogram.

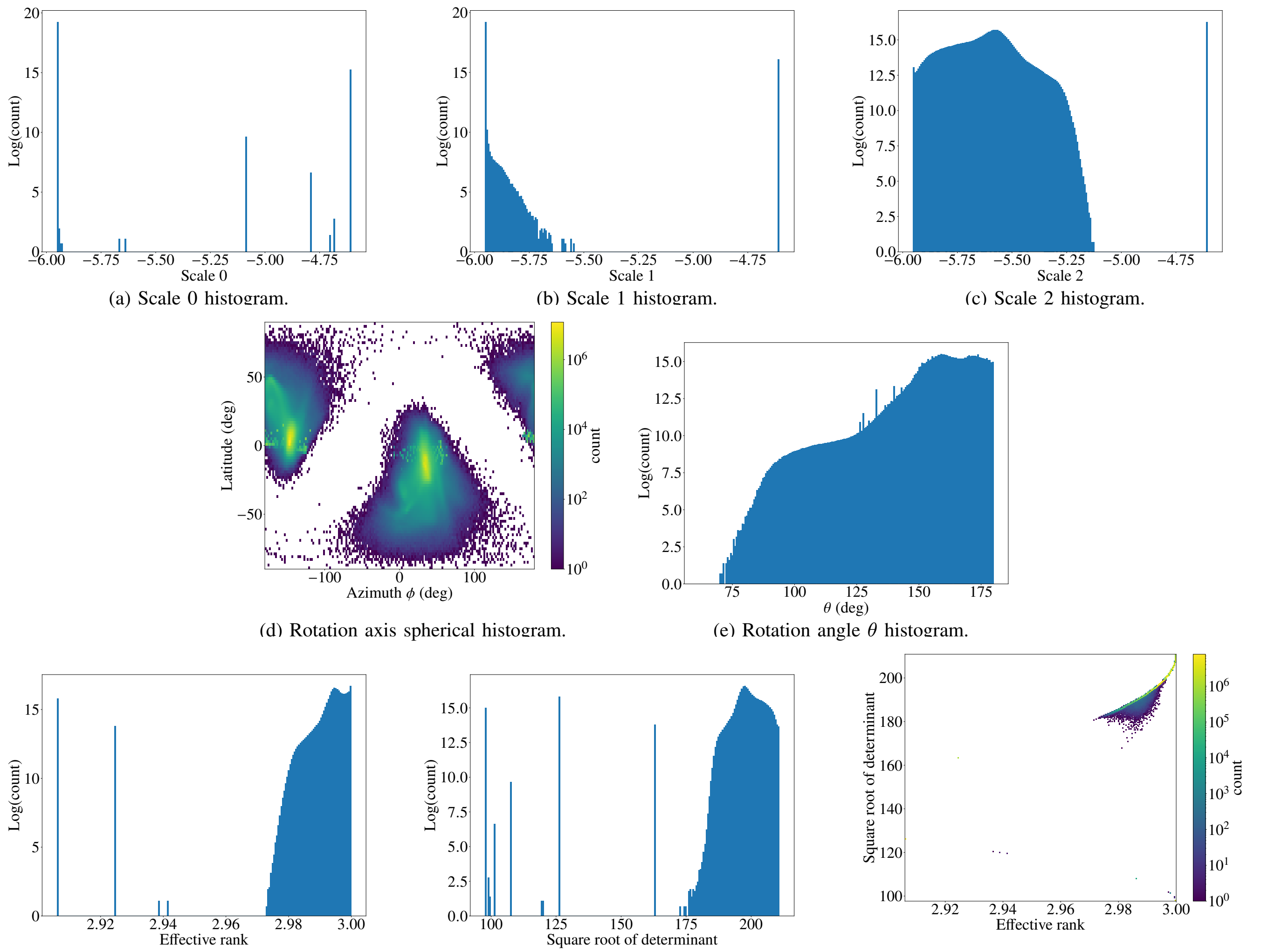
(f) Covariance matrix effective rank histogram.

(g) Square root of the covariance matrix determinant.

(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 6: Analysis of the Gaussian splats from Simone's dynamic reconstruction.

G. Thanos's Dynamic Reconstruction



(a) Scale 0 histogram.

(b) Scale 1 histogram.

(c) Scale 2 histogram.

(d) Rotation axis spherical histogram.

(e) Rotation angle θ histogram.

(f) Covariance matrix effective rank histogram.

(g) Square root of the covariance matrix determinant.

(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 7: Analysis of the Gaussian splats from Thanos's dynamic reconstruction.

H. Stimuli Watching Time

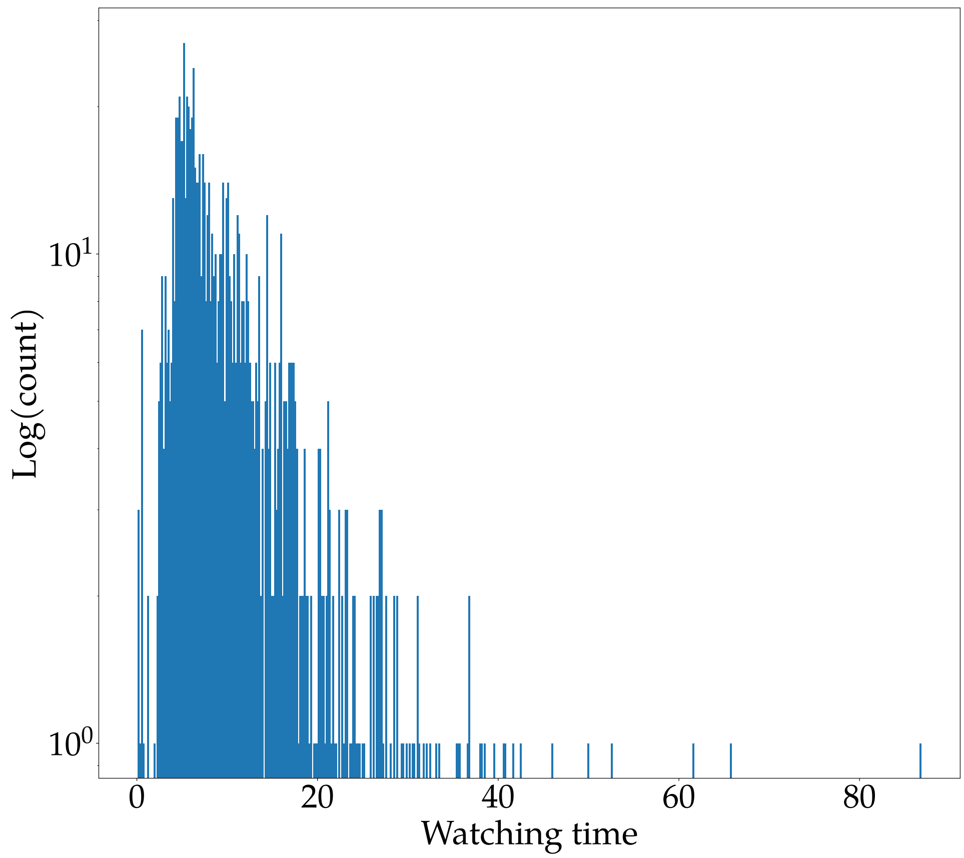
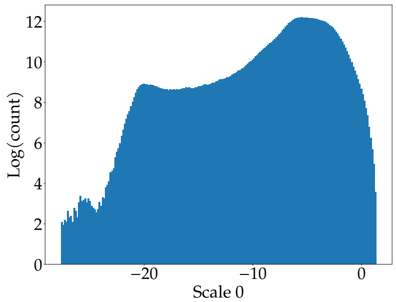


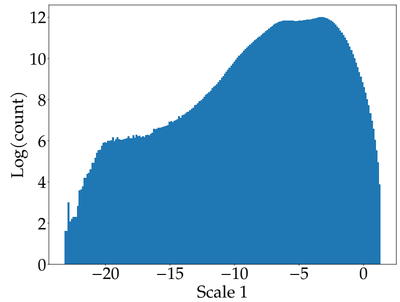
Fig. 8: Stimuli watching time histogram.

II. Gaussian Splat Analysis of Static Reconstructions

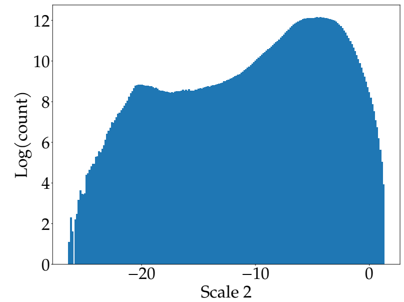
A. All Static Reconstructions



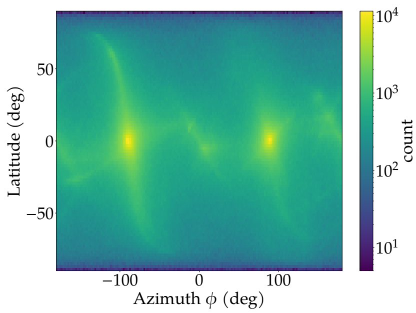
(a) Scale 0 histogram.



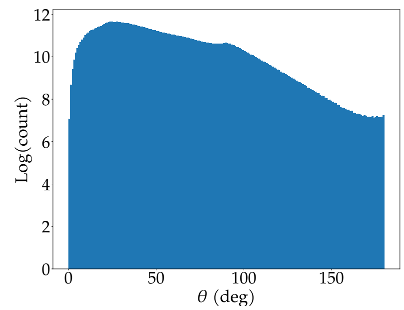
(b) Scale 1 histogram.



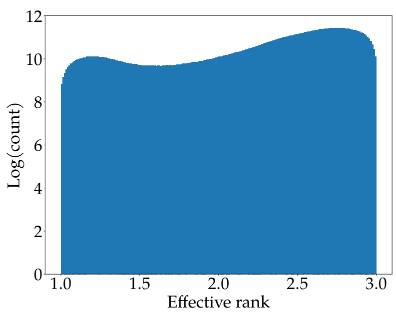
(c) Scale 2 histogram.



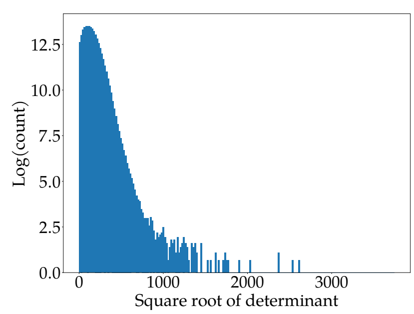
(d) Rotation axis spherical histogram.



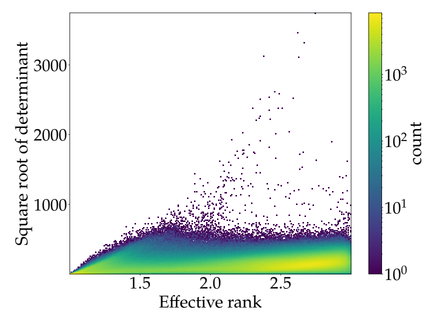
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 9: Analysis of the Gaussian splats from all static reconstructions.